FLY EMIRATES FLIGHT PERFORMANCE

Data Analysis & Business Intelligence Project Report

End-to-end pipeline: MySQL data transformation • Power BI dashboard & analysis

Dataset: U.S. Domestic Flight Performance (Jan–Mar 2015)
~1,048,575 flight records | 14 airlines | 322 airports

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1. Executive Summary

This report documents a complete data analytics project built on a large U.S. domestic flight performance dataset of roughly 1.05 million records spanning the first quarter of 2015. The objective was to take raw, messy CSV data through a full pipeline — cleaning, transformation, relational modelling, and visualisation — and turn it into an interactive Power BI dashboard that answers practical operational questions about on-time performance, delays, and cancellations.

The work was carried out across two tools, each chosen for what it does best: MySQL for the heavy cleaning and transformation of the million-row fact table and the reference tables, and Power BI for the final analytical dashboard. A three-table relational model (flights, airlines, airports) was assembled, KPIs were defined as reusable DAX measures, and a three-page dashboard was built around an overview, a routes-and-airports view, and a delay-and-cancellation analysis.

Headline results (full dataset)

On-Time Performance: ~78% of operated flights arrived within 15 minutes of schedule.

Average Arrival Delay: ~7.6 minutes across operated flights.

Cancellation Rate: ~3.9% of all scheduled flights were cancelled.

Most delay minutes came from late-aircraft and airline (carrier-controlled) causes; weather dominated cancellations.

2. Objectives & Dataset Overview

2.1 Objectives

- Clean and standardise a large, real-world flight dataset and two reference tables.
- Build a relational model linking flights to airline and airport reference data.
- Define and calculate the core operational KPIs: On-Time Performance, Average Delay, and Cancellation Rate.
- Perform exploratory analysis across airlines, airports, routes, time, and delay causes.
- Deliver an interactive, presentation-ready Power BI dashboard with clear business insights and recommendations.

2.2 Source Files

File	Description	Approx. Size
flights.csv	Fact table — one row per flight with schedule, delay, and cancellation fields	~1,048,575 rows × 31 cols
airlines.csv	Reference — IATA airline code to airline name	14 rows
airports.csv	Reference — IATA airport code, name, city, state, coordinates	322 rows

2.3 Key Fields in the Flights Table

Field group	Columns
Identity / schedule	year, month, day, day_of_week, airline, flight_number, tail_number, origin_airport, destination_airport
Times (HHMM)	scheduled_departure, departure_time, wheels_off, wheels_on, scheduled_arrival, arrival_time
Delays (minutes)	departure_delay, arrival_delay, taxi_out, taxi_in, air_time, elapsed_time
Status	diverted, cancelled, cancellation_reason
Delay causes	air_system_delay, security_delay, airline_delay, late_aircraft_delay, weather_delay

3. SQL Data Transformation (MySQL)

The full million-row transformation was executed in MySQL 8.4. This section documents each step with the representative SQL used.

3.1 Database, Table & Data Load

A database and a typed flights table were created. Time columns were declared as VARCHAR to preserve leading zeros (e.g. '0005'). The CSV was loaded from MySQL's secure Uploads folder, with empty fields converted to NULL so missing values could be filled deliberately.

```
LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.4/Uploads/flights.csv'
INTO TABLE flights
FIELDS TERMINATED BY ',' ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'   -- Windows line endings
IGNORE 1 ROWS
(@year,@month,...,@weather_delay)
SET year = NULLIF(@year,''), ... weather_delay = NULLIF(@weather_delay,'');
```

Load-stage problems solved

Error 2068 / 3948 (LOCAL INFILE disabled): enabled on both client (OPT_LOCAL_INFILE=1) and server (SET GLOBAL local_infile=1).

Error 1290 (secure-file-priv): loaded from the server's allowed Uploads folder instead.

Error 1366 ('' into integer): caused by a \r left on the last column — switching the line terminator to \r\n fixed it.

Different files had different line endings: flights = \r\n, airlines/airports = \n.

3.2 Handling Missing Values

Cancelled-flight operational fields and the five delay-cause columns were filled with 0 using COALESCE (safe-update mode disabled first).

```
SET SQL_SAFE_UPDATES = 0;
UPDATE flights SET
  departure_delay = COALESCE(departure_delay, 0),
  arrival_delay   = COALESCE(arrival_delay, 0), ... ;
UPDATE flights SET
  weather_delay = COALESCE(weather_delay, 0), ... ;
```

3.3 Decoding Coded Fields

CASE expressions replaced the Python .map() dictionaries, and an airlines lookup table provided full names via a join.

```
UPDATE flights SET cancellation_label = CASE cancellation_reason
  WHEN 'A' THEN 'Carrier' WHEN 'B' THEN 'Weather'
  WHEN 'C' THEN 'National Air System' WHEN 'D' THEN 'Security'
  ELSE 'Not Cancelled' END;

UPDATE flights SET day_name = CASE day_of_week
  WHEN 1 THEN 'Monday' ... WHEN 7 THEN 'Sunday' END;
```

3.4 Time Formatting & Date Construction

```
UPDATE flights SET sched_dep_hhmm =
  CONCAT(LEFT(LPAD(scheduled_departure,4,'0'),2), ':',
         RIGHT(LPAD(scheduled_departure,4,'0'),2));

UPDATE flights SET flight_date =
  STR_TO_DATE(CONCAT(year,'-',month,'-',day), '%Y-%m-%d');
```

Note: %Y (4-digit year) was required, not %y, to avoid an 'incorrect datetime value' error.

3.5 Delay Category

```
UPDATE flights SET departure_status = CASE
  WHEN departure_delay < 0  THEN 'Early'
  WHEN departure_delay = 0  THEN 'On Time'
  WHEN departure_delay <= 15 THEN 'Minor Delay'
  WHEN departure_delay <= 60 THEN 'Moderate Delay'
  ELSE 'Major Delay' END;
```

3.6 Duplicate Investigation — A Key Finding

An initial duplicate check using COUNT(DISTINCT) on a key that included tail_number suggested ~7,750 duplicates. However, a self-join DELETE on the unindexed million-row table ran for over 40 minutes and had to be stopped (found via SHOW PROCESSLIST and terminated with KILL).

Closer analysis revealed the count was misleading. COUNT(DISTINCT) ignores rows where any key column is NULL, and thousands of cancelled flights had a NULL tail_number. Re-checking on a meaningful flight-identity key (date + flight number + origin + destination + scheduled departure) returned only 9 genuine duplicates — a negligible 0.0009%.

```
SELECT COUNT(*) AS total,
  COUNT(*) - COUNT(DISTINCT year, month, day, flight_number,
    origin_airport, destination_airport, scheduled_departure)
  AS true_dupes
FROM flights;   -- result: total 1,048,575 | true_dupes 9
```

Lesson: choosing a deduplication key

COUNT(DISTINCT) and GROUP BY treat NULL differently — DISTINCT drops NULL rows, GROUP BY keeps them.

A key column full of NULLs (tail_number on cancelled flights) inflates apparent duplicate counts.

Always dedupe on columns that genuinely identify the entity — here, the real duplicate count was negligible and removal was unnecessary.

3.7 The Clean View

All transformations were exposed through a single analysis-ready view, joining the airlines lookup so airline names resolve automatically. Power BI reads from this view.

```
CREATE OR REPLACE VIEW flights_clean AS
SELECT f.id, f.flight_date, f.day_name,
  f.airline AS airline_code, a.AIRLINE AS airline_name,
```

```
  f.origin_airport, f.destination_airport,
  f.departure_delay, f.departure_status,
  f.arrival_delay, f.arrival_status,
  f.distance, f.cancelled, f.cancellation_label,
  f.air_system_delay, f.airline_delay, f.late_aircraft_delay,
  f.weather_delay, f.security_delay
FROM flights f
LEFT JOIN airlines a ON f.airline = a.IATA_CODE;
```

A completeness check confirmed every airline code in the fact table matched the lookup (zero unmatched rows).

4. Exploratory Analysis & KPI Queries

4.1 KPI Definitions

KPI	Definition	Denominator
On-Time Performance	% of flights arriving within 15 min of schedule	Operated (non-cancelled) flights
Average Delay	Mean arrival_delay in minutes	Operated flights
Cancellation Rate	Cancelled flights as % of scheduled	All scheduled flights

A crucial modelling rule: delay and OTP metrics filter to cancelled = 0 (a cancelled flight has no real delay and was never 'on time'), while cancellation-rate uses the full population.

4.2 Representative Queries

Top-line KPIs in a single query

```
SELECT
  COUNT(*) AS total_flights,
  ROUND(SUM(cancelled)*100.0/COUNT(*),2) AS cancellation_rate_pct,
  ROUND(SUM(CASE WHEN cancelled=0 AND arrival_delay<=15 THEN 1 ELSE 0 END)
        *100.0/SUM(CASE WHEN cancelled=0 THEN 1 ELSE 0 END),2) AS on_time_pct,
  ROUND(AVG(CASE WHEN cancelled=0 THEN arrival_delay END),1) AS avg_delay
FROM flights;
```

Average delay per airline

```
SELECT a.AIRLINE AS airline_name,
  ROUND(AVG(f.arrival_delay),1) AS avg_arrival_delay
FROM flights f LEFT JOIN airlines a ON f.airline=a.IATA_CODE
WHERE f.cancelled=0
GROUP BY a.AIRLINE ORDER BY avg_arrival_delay DESC;
```

Cancellation rate by airport

```
SELECT origin_airport,
  ROUND(SUM(cancelled)*100.0/COUNT(*),2) AS cancellation_rate_pct,
  COUNT(*) AS total_flights
FROM flights GROUP BY origin_airport
ORDER BY cancellation_rate_pct DESC;
```

Busiest routes

```
SELECT origin_airport, destination_airport, COUNT(*) AS flights,
  ROUND(AVG(CASE WHEN cancelled=0 THEN arrival_delay END),1) AS avg_delay
FROM flights GROUP BY origin_airport, destination_airport
ORDER BY flights DESC LIMIT 20;
```

Cancellation reason breakdown

```
SELECT cancellation_reason, COUNT(*) AS cancellations,
  ROUND(COUNT(*)*100.0/SUM(COUNT(*)) OVER (),1) AS pct
FROM flights WHERE cancelled=1
GROUP BY cancellation_reason ORDER BY cancellations DESC;
```

5. Power BI Data Model

A star schema was built with flights_clean as the central fact table and the reference tables as dimensions.

Relationship	Cardinality	Direction
airports[IATA_CODE] → flights_clean[origin_airport]	One-to-Many (1:*)	Single (airports → flights)
airlines[IATA_CODE] → flights_clean (optional)	One-to-Many (1:*)	Names already in view

Modelling lessons

The fact table (many rows, repeating keys) must be on the MANY side; dimensions (unique key) on the ONE side.

A 1:1 or many-to-many auto-detected relationship signals the wrong join column — the dimension must join on its unique key (IATA_CODE).

Single-direction filtering means a table visual must lead with the dimension field (airport name) to display it alongside fact measures; leading with the fact column returns blanks.

5.1 DAX Measures

```
Total Flights = COUNTROWS(flights_clean)
Cancelled Flights = CALCULATE(COUNTROWS(flights_clean), flights_clean[cancelled]=1)
Completed Flights = CALCULATE(COUNTROWS(flights_clean), flights_clean[cancelled]=0)

Cancellation Rate % = DIVIDE([Cancelled Flights],[Total Flights],0)*100

On-Time Performance % =
  VAR OnTime = CALCULATE(COUNTROWS(flights_clean),
    flights_clean[cancelled]=0, flights_clean[arrival_delay]<=15)
  RETURN DIVIDE(OnTime,[Completed Flights],0)*100

Avg Arrival Delay = CALCULATE(AVERAGE(flights_clean[arrival_delay]),
    flights_clean[cancelled]=0)
```

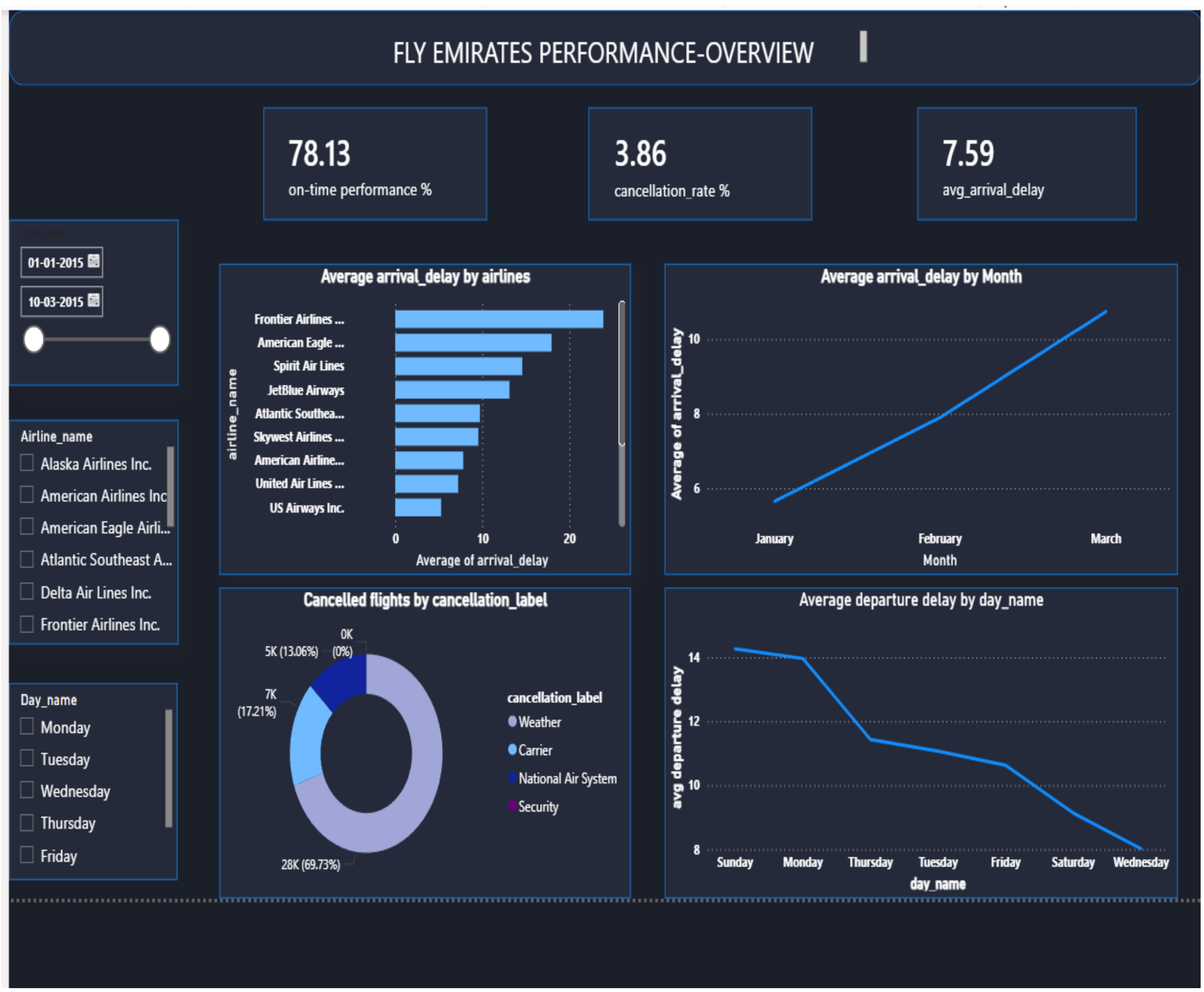
A Route calculated column (origin → destination) was added to power the busiest-routes chart, and a day_num column (from WEEKDAY of flight_date) was created so day names sort in calendar order rather than alphabetically.

6. Power BI Dashboard

A three-page dashboard was built on a dark theme, each page answering a distinct business question.

6.1 Page 1 — Performance Overview

- Three KPI cards: On-Time Performance %, Cancellation Rate %, Average Arrival Delay.
- Average arrival delay by airline (bar).
- Average arrival delay by month (line trend).
- Cancellations by reason (donut).
- Average departure delay by day of week (line, calendar-ordered).
- Slicers: airline, day, and a flight-date range.

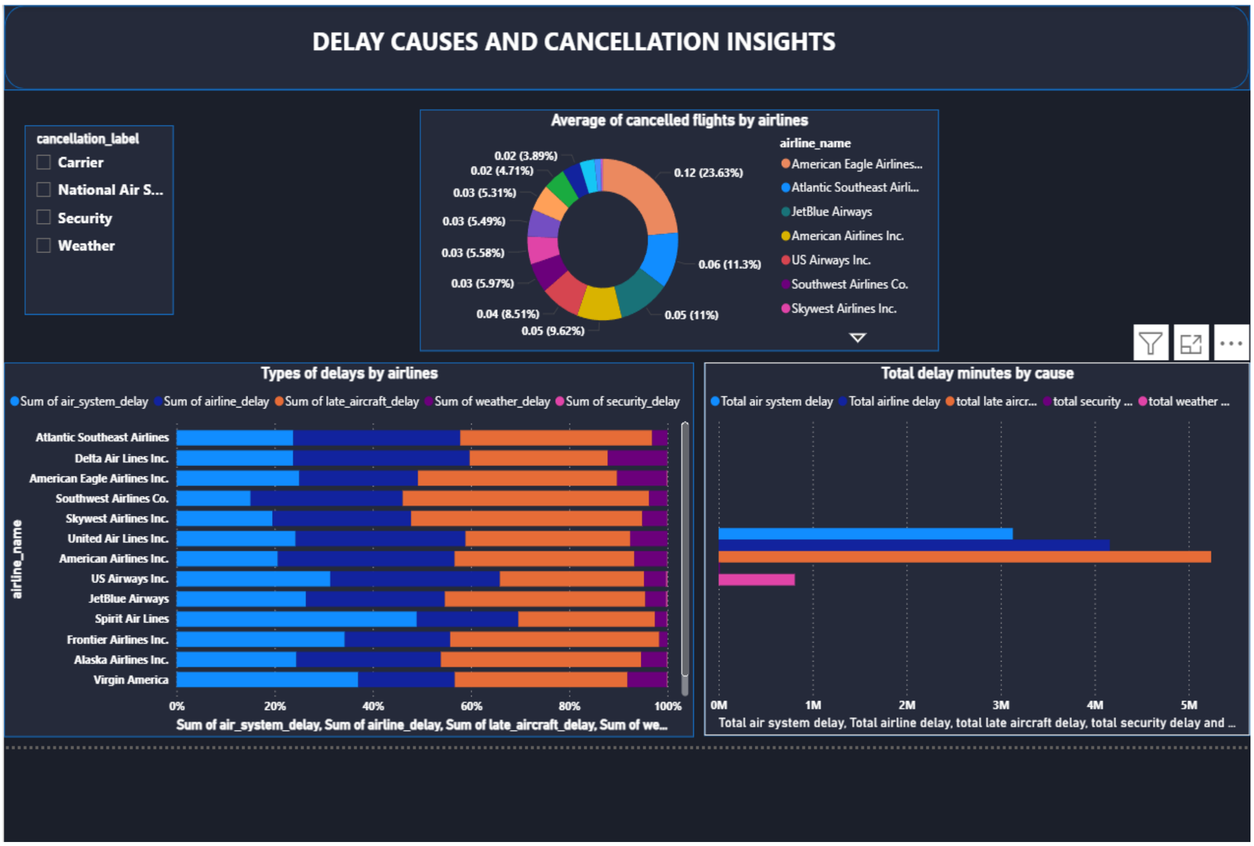


6.2 Page 2 — Routes & Airports Performance

- Airport performance table (flights, avg delay, OTP %, cancellation % per airport).
- Busiest routes (Top 15 origin → destination corridors) — e.g. JFK ↔ LAX, LAX ↔ SFO.
- Busiest origin airports (ATL, ORD, DFW, LAX, DEN leading).
- Cancellation breakdown by airline.

6.3 Page 3 — Delay Causes & Cancellation Insights

- Types of delay by airline (100% stacked bar — proportional mix of the five causes).
- Total delay minutes by cause (absolute totals — late-aircraft and airline causes dominate).
- Cancellation reason / rate by airline.
- Cancellation-reason slicer.



6.4 Interactivity

- Slicers synced across pages for a connected experience.
- Edit Interactions configured so KPI cards remain overall totals while charts cross-highlight.
- Top N filtering on route and airport charts to keep them readable.

7. Business Insights

The analysis surfaced several operationally meaningful patterns. (Figures reflect the full dataset; exact values shift when slicers are applied.)

7.1 Overall Performance

- **On-time performance sits near 78%**, meaning roughly one in five operated flights arrived more than 15 minutes late — a material reliability gap.
- **Average arrival delay is modest (~7.6 min)**, indicating most lateness is small; a minority of severe delays pulls the tail.
- **Cancellation rate is ~3.9%**, concentrated in specific causes and time periods rather than spread evenly.

7.2 Delay Causes

- **Late-aircraft and airline (carrier-controlled) delays contribute the most total delay minutes** — these are the most addressable, since they stem from operations rather than weather or airspace.
- Security delays are negligible across every airline.
- The proportional mix of causes varies by airline, so two carriers with similar average delays can have very different underlying drivers.

7.3 Cancellations

- **Weather is the single largest cancellation cause**, followed by National Air System and Carrier reasons; security cancellations are rare.
- Cancellations cluster seasonally (winter) and at weather-exposed northern hubs — they are not evenly distributed.
- Carrier-caused cancellations are the controllable subset and the fairest basis for comparing airline operational discipline.

7.4 Airlines, Routes & Timing

- Airlines differ markedly in average delay; some low-cost carriers rank worst on punctuality.
- Traffic is concentrated on a small set of high-volume corridors (JFK–LAX, LAX–SFO and similar) and major hubs (ATL, ORD, DFW).
- Delay patterns vary by day of week and rise across the observed months, consistent with seasonal and operational build-up.

8. Business Recommendations

8.1 Target Carrier-Controlled Delays First

Because late-aircraft and airline causes drive the most delay minutes and are within operational control, they offer the highest return. Tightening aircraft turnaround times, improving crew/gate scheduling, and building schedule buffers on tight connections would reduce the cascading 'late-aircraft' effect more than any weather initiative could.

8.2 Manage the Severe-Delay Tail

Since average delay is small but OTP is only ~78%, the problem is a minority of large delays. Identifying the routes, times, and aircraft rotations that generate major delays — and intervening on those specifically — would lift OTP more efficiently than broad, across-the-board measures.

8.3 Plan Around Weather-Driven Cancellations

Weather cancellations are not preventable, but they are predictable in pattern. Strengthening winter contingency planning, proactive rebooking, and capacity buffers at weather-exposed hubs during peak-risk months would reduce passenger disruption and protect schedule integrity.

8.4 Benchmark Airlines on Controllable Metrics

Compare carriers on carrier-caused cancellations and airline/late-aircraft delay share, not raw totals. This isolates operational discipline from factors outside an airline's control and gives a fairer, more actionable performance league table.

8.5 Focus Improvement on High-Volume Corridors

Because traffic concentrates on a few routes and hubs, performance gains there have outsized network impact. Prioritising reliability investment on the busiest corridors and at the largest origin airports yields the broadest passenger benefit per unit of effort.

Summary recommendation

Roughly the majority of improvable delay is carrier-controlled. Concentrating operational effort on aircraft turnaround, the severe-delay tail, and the busiest hubs — while planning defensively around weather — is the most cost-effective path to higher on-time performance and fewer disruptions.

9. Tools, Pipeline & Conclusion

9.1 Pipeline Summary

Stage	Tool	Purpose
Data transformation	MySQL 8.4	Load, clean, decode, deduplicate-check, and model ~1M rows; build the clean view
Modelling & visualisation	Power BI	Star schema, DAX KPIs, interactive 3-page dashboard

9.2 Conclusion

This project delivered a complete, reproducible analytics pipeline from raw CSV to interactive dashboard. Beyond the deliverable itself, the process reinforced several durable data-analysis principles: the importance of preserving missingness rather than silently coercing it, the difference between alphabetical and semantic ordering, the critical role of choosing a correct entity key when deduplicating, and the need to separate controllable from uncontrollable factors when drawing business conclusions.

The resulting dashboard answers the three questions a flight-operations stakeholder cares about most — how are we performing, where is the traffic, and what is going wrong — and the recommendations point clearly at the carrier-controlled levers that would most improve on-time performance.

End of Report